

ECE 447

Lesson 30

Probability, Part 1



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **HW6.** Was due Lesson 28 (27 Oct).
 - **Lab 6.** PDF due Lesson 31 (Wed, 4 Nov) to Gradescope.
 - **HW7.** Due Lesson 32 (Fri, 6 Nov) to Gradescope.
 - **HW8.** Assigned Lesson 31 (Wed, 4 Nov); due Lesson 34 (Fri, 13 Nov).
 - **GR2** is Lesson 35 (Tue, 17 Nov), covering Lessons 18-32 (Chapters 5-8). Review day is Lesson 34 (Fri, 13 Nov).

REVIEW

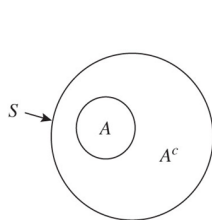
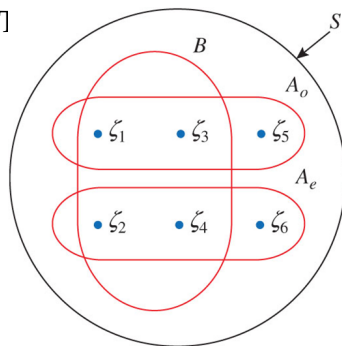
- A binary FSK signal uses carrier frequencies f_A and f_B . The baseband signal uses full-width polar signaling. Estimate the transmitted bandwidth of the FSK signal using Carson's Rule.

WHY PROBABILITY IN DIGITAL COMMUNICATIONS?

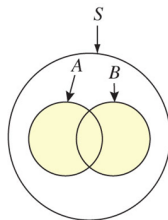
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- What did you learn in Math 356 and ECE 346?

PROBABILITY TERMS & CONCEPT

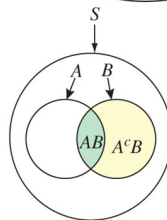
- Experiment, outcomes, events
- Sample space S , elements, complement
- Union $\cup$, intersection $\cap$
- Disjoint or mutually exclusive
- Conditional probability, e.g., $P(B|A)$
- Bayes' rule $P(A|B) = \frac{P(A)P(B|A)}{P(A)}$

 A^c

(a)

 $A \cup B = B \cup A$

(b)

 AB and A^cB

(c)

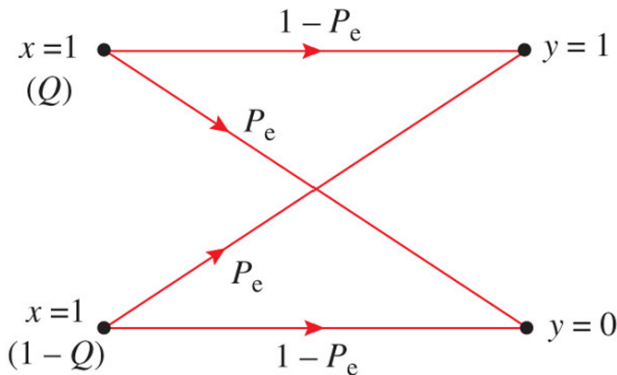
PROBABILITY TERMS & CONCEPTS (CONT'D)

- Independent: iff $P(A \cap B) = P(A)P(B)$ (NOT the same as mutually exclusive)
- Bernoulli trials
 - $P(k \text{ successes in } n \text{ trials}) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$
 - p is the probability of success of a single trial
 - Example: BSC w/ $P_e = 10^{-3}$. What is the probability that a nibble of a byte is incorrect?
- Example: Repetition codes; see Example 7.9 in textbook
- Total Probability Theorem: $P(B) = \sum_{i=1}^n P(B|A_i)P(A_i)$
- Prior and posterior probabilities, $P(A_i)$ and $P(A_i|B)$
 - Find posterior probability through combination of Bayes' rule and Total Probability Theorem
 - $P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$

RANDOM VARIABLES

- Maps events to real numbers
- Notation
 - $P_x(x_i)$: x is RV, x_i is value RV takes; "Probability of $x = x_i$ "
- Can have multiple RVs or *joint probabilities*
 - $P_{xy}(x_i, y_j)$: "Probability that $x = x_i$ and $y = y_j$ "
 - If x and y are independent, $P_{xy}(x_i, y_j) = P_x(x_i)P_y(y_j)$
- Conditional probabilities
 - Probability of $x = x_i$ given $y = y_j \rightarrow P_{x|y}(x_i|y_j)$
- Marginal probabilities
 - $P_y(y_j) = \sum_i P_{xy}(x_i, y_j) = \sum_i P_{x|y}(x_i|y_j)P_y(y_j)$
 - $P_x(x_i) = \sum_j P_{xy}(x_i, y_j)$

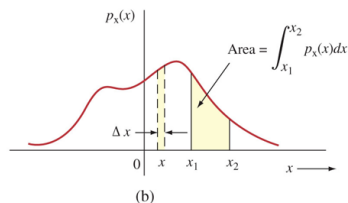
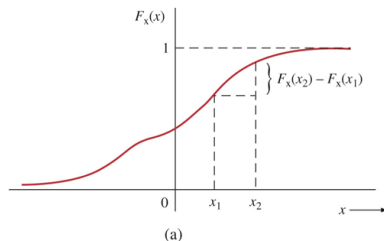
EXAMPLE: BINARY SYMMETRIC CHANNEL (BSC)



x is the transmitter. The probability of transmitting a 1 is Q . The probability of a bit error due to the channel is P_e . Find the probability of receiving a 1 or 0 at the receiver y .

CDF AND PDF

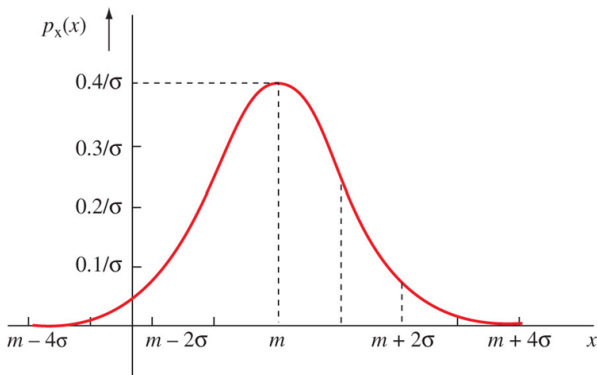
- CDF: $F_X(x) = P(X \leq x)$
 - $F_X(x) \geq 0$
 - $F_X(\infty) = 1$
 - $F_X(-\infty) = 0$
 - $F_X(x_1) \leq F_X(x_2)$ for $x_1 \leq x_2$
- PDF: $p_X(x) = \frac{dF_X(x)}{dx}$
 - $\int_{-\infty}^{\infty} p_X(x) dx = 1$
 - $P(x_1 < x \leq x_2) = \int_{x_1}^{x_2} p_X(x) dx$



GAUSSIAN RV

$$p_x(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-m)^2/(2\sigma^2)}$$

- Cover this next time...



BACKUP: REPEATED TRIALS (LATHI 7.1-15)

- A receiver detects binary pulses with error probability P_e . Out of 100 received digits, what is the probability that **no more than four** are in error?
- Each digit is an independent trial, so the error count k is binomial with $n = 100, p = P_e$
- “No more than four” means $k \in \{0, 1, 2, 3, 4\}$ — mutually exclusive, so the probabilities

- $$P(k \leq 4) = \sum_{k=0}^4 \binom{100}{k} P_e^k (1 - P_e)^{100-k}$$
- No value is given for P_e , so this expression *is* the answer. How many terms is it? _____
- Poisson sanity check with $\lambda = nP_e$: at $P_e = 0.01$ it gives 0.9963 against the exact 0.9966